

Transplant Medicine — Immunosuppression & Post-Transplant Complications

Balancing rejection vs. infection/cancer/toxicity · induction vs. maintenance · complications by time from transplant. IM facilitator reference.

The Core Balance

More immunosuppression → less rejection, but more **infection, malignancy, drug toxicity**. Therapy is titrated to each organ's rejection risk and time from transplant.

Allogenicity (rejection risk): Lung > heart/kidney > liver. Lung ~50% reject by 5 yr (often on all 3 classes for life); liver least (may be on CNI alone).

Induction vs. Maintenance

Phase	When	Agents
Induction	At transplant	Anti-lymphocyte Ab (rATG, alemtuzumab) or anti-IL-2R (basiliximab) + high-dose steroids. <i>OKT3 discontinued 2010.</i>
Maintenance	Long-term	CNI + antimetabolite ± prednisone

Maintenance classes: CNI (tacrolimus, cyclosporine) · antimetabolite (mycophenolate, azathioprine) · steroid (prednisone) · mTOR inhibitor (sirolimus, everolimus).

Complication Timeline

Time	Infection	Other
<1 mo	Nosocomial; donor/recipient-derived	Surgical (leaks, kinks, fistulas, stenoses)
1–6 mo	Reactivation/OIs: CMV, BK, HSV/VZV, TB, HBV; PJP, Crypto, Toxo (↓ w/ prophylaxis)	Acute rejection peak; PTLN (EBV) can begin
>6 mo	Community-acquired » OIs; late viral	Chronic rejection; late solid malignancies

Rejection

- **Acute cellular** (T-cell): biopsy = lymphocytic infiltrate → pulse methylpred 1 g ×3 d ± anti-lymphocyte Ab.
- **Acute humoral** (Ab-mediated): biopsy C4d / donor-specific Abs → ↑ immunosuppression, IVIg, plasmapheresis, rituximab, bortezomib, eculizumab.
- **Chronic:** lung = bronchiolitis obliterans (obstructive, ↓FEV₁; azithromycin 3×/wk); kidney = chronic allograft dysfunction (↑Cr, proteinuria).
- **First** acute rejection >6 mo unlikely if adherent (~25% have an acute episode overall).

Drug Toxicities & Malignancy

- **CNI:** nephro/neurotoxicity, ↑K, ↑uric acid, metabolic syndrome.
- **Antimetabolite:** marrow suppression, GI (colitis). **Steroid:** metabolic syndrome.
- **Cancer ~2×** general population — virally driven (PTLD/EBV early; skin/anogenital HPV, liver late).

Newer Context (cite before teaching)

- **Belatacept** (Nulojix, 2011): CD28 costimulation blocker, CNI-sparing; better long-term GFR. **Contraindicated if EBV-seronegative** (PTLD/PML).
- **Letermovir** (2023): CMV prophylaxis in high-risk D+/R– kidney Tx; less myelosuppression than valganciclovir.
- **Maribavir** (2021): refractory/resistant CMV.

High-Yield Pearls

- A NEW fever in a transplant patient = think infection AND rejection; both can present with malaise and graft dysfunction.
- Match the bug to the clock: nosocomial <1 mo, opportunistic/reactivation 1